Friedrich-Alexander-Universität
Erlangen-Nürnberg

Advanced Laboratory Course

Photovoltaic: Function and electrical Characterization of Solar Cells [EN]

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Group ♣

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1 Preparation

1.1 Abstract

In this experiment we measure the main parameters of two solar cells related to their volt-ampere characteristic. We make measurements both without illuminating them and with illuminating them.

Below we briefly describe the corresponding theory and our experiment, mostly following [1] and [2].

1.2 Theoretical basics

1.2.1 Band Model

An electron with energy E and wave-function Ψ in a periodic crystal structure giving potential $U(\vec{r})$ is described by some Schrodinger equation:

$$\nabla^2\Psi + \frac{2m}{\hbar^2}(E - U(\vec{r}))\Psi = 0 \tag{1}$$

This leads to the spectrum $E(p)$, giving dependence of energy E on momentum p. It is known that this spectrum is given by Fig 1 and can be approximated by two parabolas.

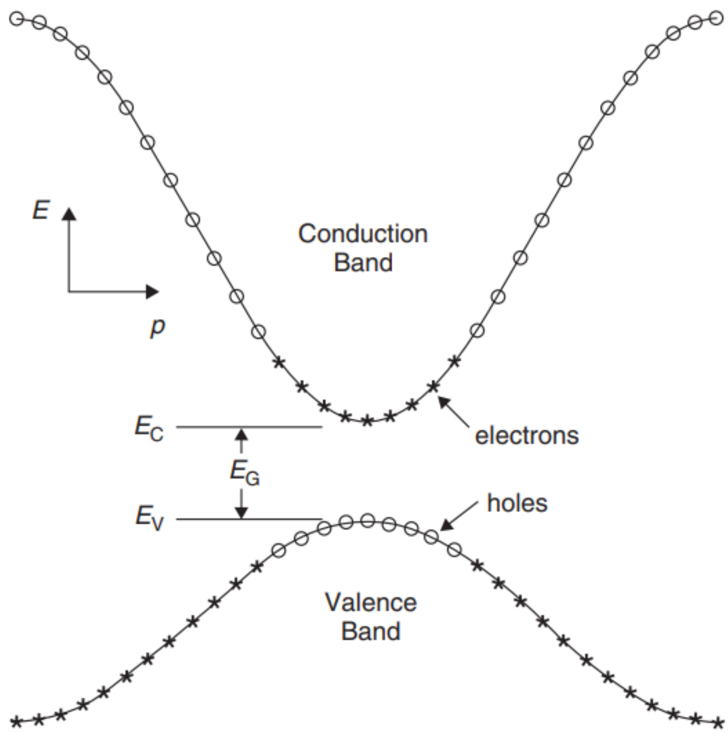

Figure 1: Spectrum in periodic crystal structure.

The second order coefficients of this parabolas are called effective masses. Electrons from the bottom (valence) band can go to the top (conduction) band. They are usually called just electrons (labeled n), with an effective mass m_n^*, while corresponding left empty states from the bottom usually called holes (labeled p), with the mass m_p^*. Holes are interpreted as particles themselves, having positive charge $+e$.

1.2.2 Densities of states

One of the most fundamental quantities in quantum solid state physics is density of states $g(E) = \frac{\partial n(E)}{\partial E}$, where number of states $n(E)$ per volume with energy E differentiating with respect to it. It is known that for the conduction band we have:

$$g_{\mathrm{C}}(E) = \frac{m_n^* \sqrt{2m_n^* \left(E - E_{\mathrm{C}}\right)}}{\pi^2 \hbar^3} \ \mathrm{cm}^{-3}\mathrm{eV}^{-1} \tag{2}$$

While for the valence band:

$$g_{\mathrm{V}}(E) = \frac{m_p^* \sqrt{2m_p^* \left(E_{\mathrm{V}} - E\right)}}{\pi^2 \hbar^3} \ \mathrm{cm}^{-3}\mathrm{eV}^{-1} \tag{3}$$

Taking in account the temperature, one need to useFermi distribution, which describes average number of electrons as

$$f(E) = \frac{1}{1 + \mathrm{e}^{(E - E_{\mathrm{F}})/kT}}, \tag{4}$$

where E_F is Fermi energy. It is important to note that holes are then described by $1 - f(E)$, resulting in formula of the same kind but with replaced $E_F \to -E_F$ and $E \to -E$. This leads to the following concentrationes of, respectively, electron and holes:

$$n_o = \int_{E_{\mathrm{C}}}^{\infty} g_{\mathrm{C}}(E) f(E) \mathrm{d}E = \frac{2N_{\mathrm{C}}}{\sqrt{\pi}} F_{1/2}\left(\left(E_{\mathrm{F}} - E_{\mathrm{C}}\right)/kT\right) \tag{5}$$

$$p_o = \int_{-\infty}^{E_{\mathrm{V}}} g_{\mathrm{V}}(E)[1 - f(E)] \mathrm{d}E = \frac{2N_{\mathrm{V}}}{\sqrt{\pi}} F_{1/2}\left(\left(E_{\mathrm{V}} - E_{\mathrm{F}}\right)/kT\right) \tag{6}$$

$$F_{1/2}(\xi) = \int_0^{\infty} \frac{\sqrt{\xi'} \mathrm{d}\xi'}{1 + \mathrm{e}^{\xi' - \xi}} \tag{7}$$

1.2.3 Doping

Conducting properties of the semiconductor are variable on practice using so-call doping, which means adding other materials which have their own carries of the charge. There two types of the doping: n-type, where additional carries are electrons, and p-type, where additional carries are holes. Thus, the current can flow mainly with holes or electrons, which are called majority carriers, while secondary ones are minority.

1.2.4 pn-junction diode

Consider coupled p-doped semiconductor on one side and n-doped semiconductor on the other side. Due to having different densities of electrons and holes they create diffusion zone, where, as well known in general, the concentration of the neighbor's majority carrier exponentially decreases. In addition, without electrical equilibrium, the existing electrical field leads to so-called drift current.

Thus, there are two currents, one diffusion current and one drift current, all them involves moving holes and electrons. Considering statistical current situation neglecting displacement current from Maxwell equations, total current is simply given by:

$$I = I_{diff} + I_{drift} = I_{diff\,p} + I_{diff\,n} + I_{drift\,p} + I_{drift\,n} \tag{8}$$

The equilibrium between p-type and n-type semiconductors leads to the well-known band bending structure, shown on Figure 2. Consider U-I characteristics of the diod. Roughly speaking,

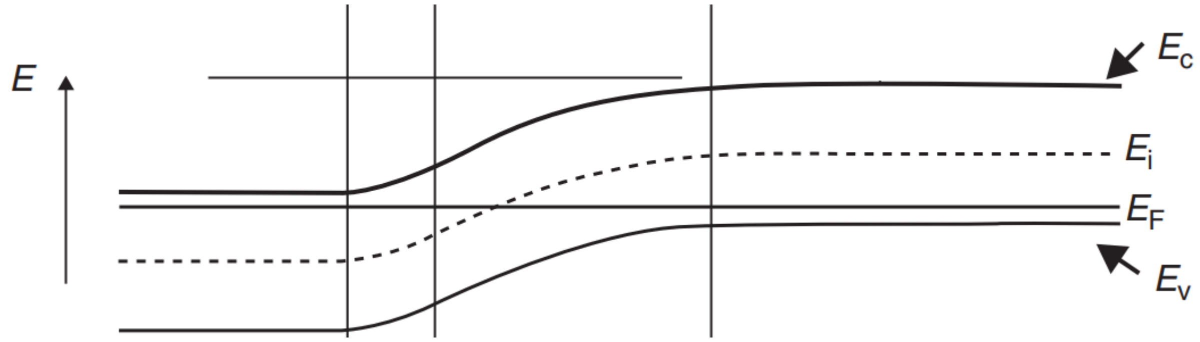

Figure 2: Band bending for connection of p-type and n-type superconductors. The main thing to note is that the fermi level becomes the same, and the energy levels shift, remaining unchanged relative to it. Here E_i is Fermi energy in an intrinsic, without dopping, semiconductor in thermal equilibrium.

we have applied voltage U and barrier voltage between the types U_B, which leads to:

$$I_{diff} \sim e^{\frac{(U-U_B)e}{kT}} \tag{9}$$

Using also the fact that at $U = 0$ diffusion ad drift current cancels each other due to the equilibrium, so $I_{drift} \sim e^{\frac{-eU_B}{kT}}$, we finally get the total current:

$$I(U) \sim \left(e^{\frac{Ue}{kT}} - 1\right) \tag{10}$$

However, due to several reasons, real diod is not ideally described by this simple consideration, and one more factor A (so-called diod factor, showing this deviation) need to be introduced in U-I characteristic of the diod:

$$I(U) = I_0 \left(e^{\frac{Ue}{AkT}} - 1\right) \tag{11}$$

1.2.5 Light absorption

The main effect, on which photovoltaic is based, is the absorption of light leading to creation of an electron-hole pair.

Consider photon with frequency ν falling on the semiconductor. If $h\nu$ is bigger than E_G, this could lead to absorption and transition of the (fundamental) electron from valence band to the conduction band, see Figure 3. Due to conservation of the total energy and the momentum of that electron p, we have:

$$h\nu = E_2 - E_1 \tag{12}$$

$$E_V - E_1 = \frac{p^2}{2m_p^*} \tag{13}$$

$$E_2 - E_C = \frac{p^2}{2m_n^*} \tag{14}$$

This leads to:

$$hv - E_G = \frac{p^2}{2}\left(\frac{1}{m_n^*} + \frac{1}{m_p^*}\right) \tag{15}$$

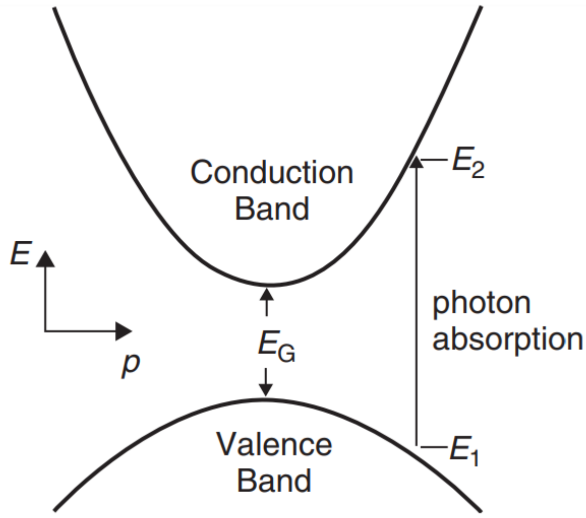

Figure 3: Absorption of the photon leading to the transition.

Thus, this result can be interpreted in terms of solid state terminology as creation of an electron and a hole, both with momentum p.

1.2.6 Solar cells

The solar cell is basically pn-diod which is illuminated by a light. The falling light reaches the diffusion zone between p-type and n-type semiconductors and creates holes and electrons. Light destroys thermal equilibrium and holes ca move to the p-type as electrons can move to the n-type. Due to the existing potential difference, it is impossible for them to go back. Thus, solar cell generates it's own voltage leading to the current in the electrical circuit.

U-I characteristics of the illuminated solar cell is generalized to the following form:

$$I(U) = I_0\left(e^{\frac{Ue}{AkT}} - 1\right) - I_{photo} \tag{16}$$

The example of the corresponding graph can be viewed from Figure 4.

There are several usual characteristic of any solar cell:

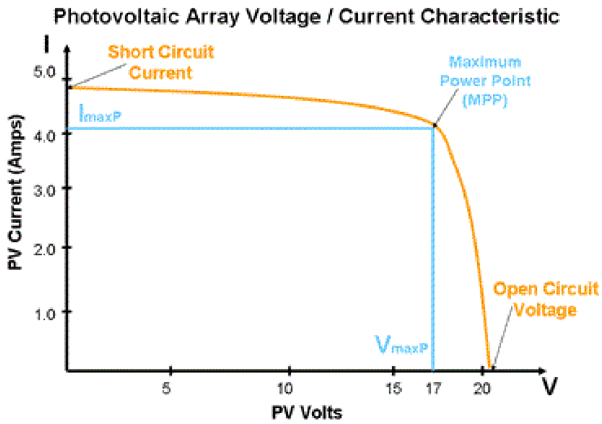

Figure 4: Example of U-I characteristic of the solar cell.

Open circuit voltage U_0: $I(U_0) = 0$.
Short circuit current I_k: $I_k = I(U = 0)$
Point of maximum power P_{max}.
Efficiency η: $\eta = \frac{P_{max}}{P_{radiation}}$
Optimum load resistance R_{opt}: resistance for which the solar sell work at it's maximum power.
Filling factor F: $F = \frac{P_{max}}{U_0 I_k}$

There two the most important type of solar cells:the crystalline and the amorphous solar cell. Although they generally work in the same way, it is important to note some differences. As follows from the name, they are distinguished by the presence or absence of a crystalline structure in the semiconductor material. Although they generally work in the same way, it is important to note some differences. For example, the diffusion length of the formed particles in the crystalline cells is much greater than in the amorphous cells. So it comes to use additional material between n and p types in the construction for the second ones. However, the absorption characteristics of the amorphous material are better.

1.3 The experiment

1.3.1 Experimental setup

In this experiment, we will use solar cells (crystalline and the amorphous) by connecting them to various electrical circuits, using voltmeters, ammeters, resistors, a voltage source. In addition we use a dark box, a silicon reference diode, a Halogen light bulb, KG 2 filter. Below we shortly describe parts of the experiment.

1.3.2 Dark curve measurement

In the first part we measure the U-I curve of the two solar cells without any illumination. We connect them according to the picture with $R_p = 100\Omega$.

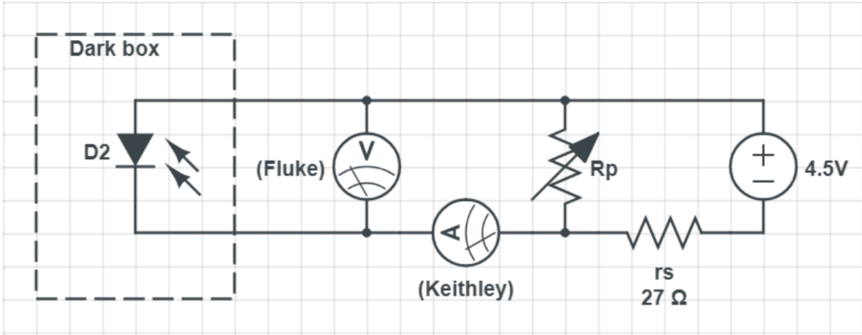

Figure 5: The solar cell (D3) connection diagram. Here A is ammeter, B is voltmeter, Rp1 ad Rp2 are resistors

1.3.3 Measurement under illumination

The values we measure in the presence of illumination correspond to a different U-I graph quadrant than those in the preceding section. Using a 100W halogen lamp and a KG 2 filter, which blocks out the majority of IR intensity, creates light that‘s spectrally similar to that of the sun. Calibration of the setup is the initial stage. To do this, we adjust the distance between the silicon detector and the lamp while using the silicon reference diode to measure the light intensity at least 15 separate times. The diode's 118 W/cm2 A calibration factor is only valid when the KG 2 filter used. We pick two points on the positioning rail, one with a low intensity and one with a high intensity (for instance, 2.5mW/cm^2 and 10mW/cm^2), whose light intensities are now known. While the cells are illuminated, measure the U-I curve of each cell at both of these positions (4 curves altogether). The calibration diode (34mm), crystalline (7mm), and amorphous (10mm) cells have varying distances from the center of the mounting pole on the rail to the photo-sensitive surface.

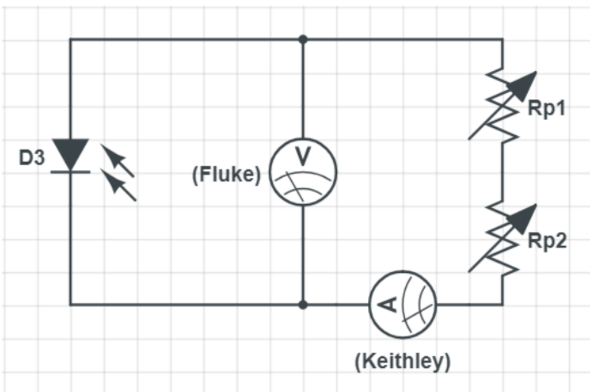

Figure 6: The solar cell (D3) connection diagram. Here A is ammeter, B is voltmeter, Rp1 ad Rp2 are resistors

1.3.4 Open circuit voltage and short circuit current

At 15 different distances from the filter, measure the short circuit current Ik and the open circuit voltage U0. Make a note of the corresponding intensity that the calibration diode measured. The measurement could be affected by the cells heating up.

1.3.5 Field experiment

Finally, we take the solar cells out to some sunny place and measure the characteristics U_0 and I_k. We also take in account the time of the day and the angle between the cells and the ground. Using the calibration diod, we also measure the sunlight intensity.

1.3.6 Experimental notes

Following [1], it is important to note:

(4.2a) According to the U-I characteristics, derived before, we don't need external voltage source to have nonzero voltage and current due to the I_{photo} term.
(4.2b) It is important to wait at least 20 after turning on the bulb min to make sure that created temperature is homogeneous.
(4.2c) The known calibration factor is related to the system using KG2. Using other filters or without filters will change the properties, so the information will not be correct.
(4.3a) Heating up of the cells leads to increasing of the radiation which can influences the measurement.
(4.3b) Measuring the time of the day and data is important since the sunlight varies in composition and intensity depending on the angle of passage through the atmosphere because its path is different. The angle of radiation incident on the cells is also important, since the flux depends on it.

References

[1] Photovoltaic: Function and electric characteristics of solar cells made from crystalline and amorphous silicon (B33), Studon 2019.

[2] https://ee.sharif.edu/ sarvari/25798/solarcell-ch3.pdf

[3] https://www.electrical4u.com/solar-cell

[4] H Kang, Crystalline Silicon vs. Amorphous Silicon: the Significance of Structural Differences in Photovoltaic Applications, IOP Conference Series: Earth and Environmental Science 2021. URL: https://iopscience.iop.org/article/10.1088/1755-1315/726/1/012001